7.5.5 Treatment and Prophylaxis

Penicillin – G is the drug of choice. In penicillin allergic cases, chloramphenicol is recommended.

- Monovalent and polyvalent vaccines (capsular polysaccharide) induce good immunity in older children and adults.
- Conjugate vaccines are used for children below the age of 2 years.

7.6 *Corynebacterium Diphtheriae*

Several species of the genus *Corynebacterium* are normal flora of skin, upper respiratory tract (URT), urogenital and intestinal tract. The most important member of the genus is *C. diphtheriae* the causative agent of diphtheria, a localized inflammation of the throat with greyish white pseudomembrane and a generalized toxemia due to the secretion and dissemination of a highly potent toxin.

The name *Corynebacterium diphtheria* is derived from **Greek word 'Coryne'** – "Club shaped swellings" or"Knotted rod" **'Diphthera'** – Leather.

7.6.1 Morphology

- They are Gram positive slender rods, pleomorphic club shape or coryneform bacterium Non – motile, non – sporing and non – capsulated (Figure 7.9a & b).
- The bacilli are arranged in a characteristic fashion in angular fashion resembling the letters V or L. This has been called Chinese letter or cuneiform arrangement (Figure 7.10).
- They are club shaped due to the presence of metachromatic granules at one or both ends. These granules are composed of polymetaphosphates and represent energy storage depots.

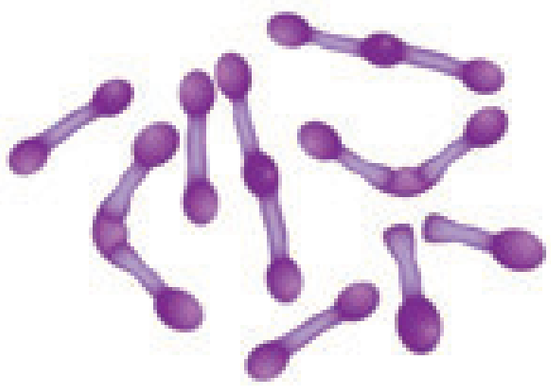


Figure 7.10: Gram staining of *Corynebacterium diphtheriae*

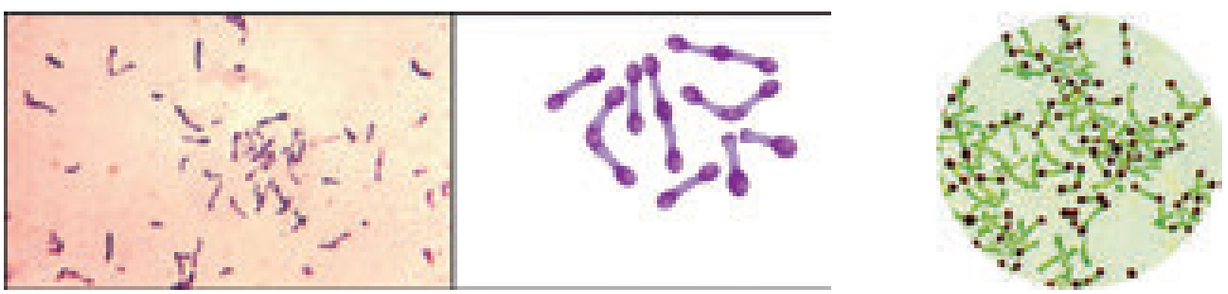


Figure 7.9: (a) Gram staining of *Corynebacterium diphtheriae* **(b)** Albert's staining showing metachromatic granules

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